

10A Practice Solutions

Q1

$$\begin{aligned} \text{a) } (x^2 y^3)^4 \\ = x^{2 \times 4} y^{3 \times 4} \\ = x^8 y^{12} \end{aligned}$$

$$\begin{aligned} \text{b) } \left(\frac{2xy^4}{xy} \right)^3 \\ = 2^3 y^3 \\ = 8y^3 \end{aligned}$$

$$\begin{aligned} \text{Q2 a) } \frac{x^4 y}{y^3 z} \times \frac{\cancel{x^2} y^2}{\cancel{x^3}} \\ = \frac{x^4 \cancel{y}}{\cancel{y^3} z} \times \frac{\cancel{y^2}}{\cancel{x^3}} \\ = x^3 \end{aligned}$$

$$\begin{aligned} \text{b) } \left(\frac{x^2}{y^{-1}} \right)^{-2} \\ = (x^2 y)^{-2} \\ = \frac{1}{x^4 y^2} \end{aligned}$$

$$\begin{aligned} \text{Q3 a) } \sqrt[5]{x^2} \\ = x^{\frac{2}{5}} \end{aligned}$$

$$\begin{aligned} \text{b) } (\sqrt{y})^3 \\ = y^{\frac{3}{2}} \end{aligned}$$

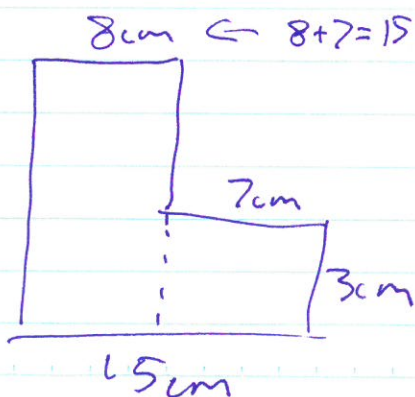
$$\text{Q4 } 8^{\frac{1}{3}} = 2$$

$$\begin{aligned} \text{Q5. a) } 4\sqrt{7} + 9\sqrt{7} \\ = \underline{13\sqrt{7}} \end{aligned}$$

$$\begin{aligned} \text{b) } 5\sqrt{3} \times 3\sqrt{12} \\ = 15\sqrt{3 \times 12} \\ = 15\sqrt{36} \\ = 15 \times 6 \\ = \underline{90} \end{aligned}$$

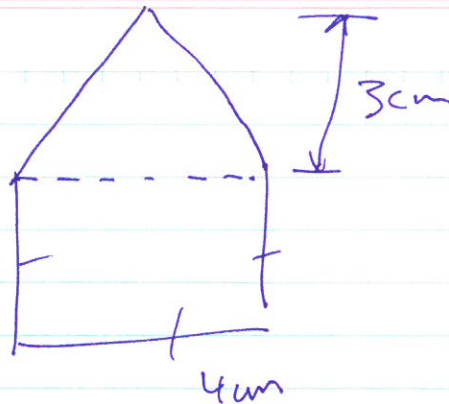
Q6 a)

10cm



$$\begin{aligned} A &= 7 \times 3 + 10 \times 8 \\ &= 21 + 80 \\ &= \underline{101 \text{ cm}^2} \end{aligned}$$

Q6 b)



Triangle Square

$$A = \frac{1}{2} \times 4 \times 3 + 4 \times 4$$

$$= 6 + 16$$

$$= \underline{22 \text{ cm}^2}$$

Q7 a)



$$V = 4 \times 4 \times 7$$

$$= 112 \text{ cm}^3$$

$$SA = 4 \times 4 \times 2 \quad 16$$

$$7 \times 4 \times 4 \quad 56$$

$$+ \underline{72 \text{ cm}^2}$$

b)



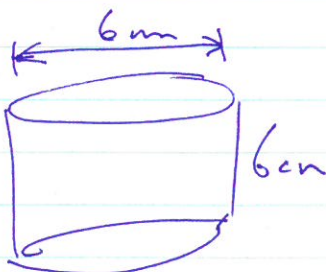
$$V = \frac{4}{3} \pi r^3$$

$$= 268.1 \text{ cm}^3$$

$$SA = 4 \pi r^2$$

$$= 201.06 \text{ cm}^2$$

c)



$$V = \pi r^2 h$$

$$= \pi \times 3^2 \times 6$$

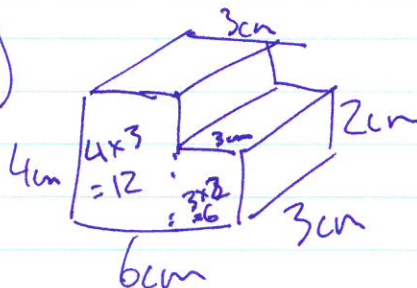
$$= \underline{169.65 \text{ m}^3}$$

$$SA = 2 \pi r^2 + 2 \pi r h$$

$$= 2 \times \pi \times 3^2 + 2 \times \pi \times 3 \times 6$$

$$= \underline{169.65 \text{ m}^2}$$

d)



$$V = \text{Area of face} \times \text{height}$$

$$= 18 \times 3$$

$$= \underline{54 \text{ cm}^3}$$

$$SA = (18 \times 2) + (4 \times 3 \times 2) + (6 \times 3 \times 2)$$

$$= 36 + 24 + 36$$

$$= \underline{96 \text{ cm}^2}$$

$$\begin{array}{r} \text{Q 8 a)} \quad x+y=5 \\ + \quad x-y=1 \\ \hline \end{array}$$

$$2x+0=6$$

$$x=3$$

$$x+y=5$$

$$y=2$$

$$\therefore x=3, y=2$$

$$(3, 2)$$

$$\begin{array}{r} \text{b)} \quad y=3x \\ 2x-y=-6 \end{array}$$

$$2x-3x=-6$$

$$-1x=-6$$

$$x=6$$

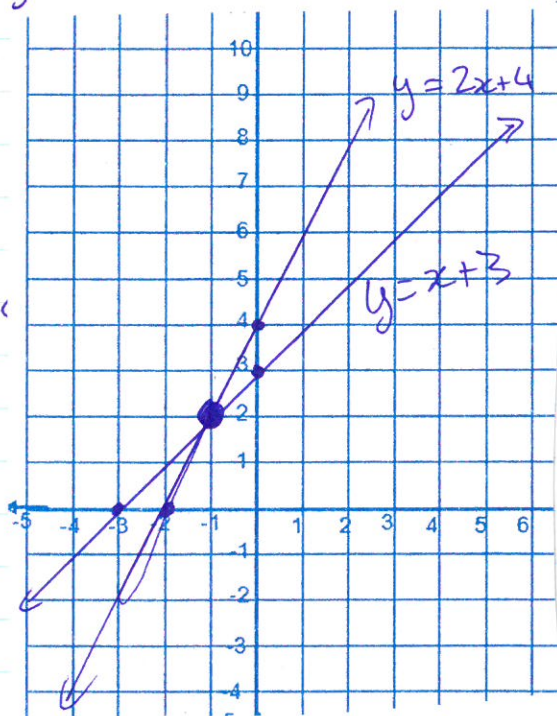
$$y=3 \times 6$$

$$=18$$

$$\therefore x=6, y=18$$

$$(6, 18)$$

$$\text{c)} \quad y=x+3 \text{ and } y=2x+4$$



$$(-1, 2)$$

↑
solution is point of intersection

$$\begin{array}{r} \text{Q 9 a)} \quad (x+2)(x-2) \\ = x^2 - 4 \\ \hline \end{array}$$

$$\text{b)}$$

$$\begin{array}{r} (2x+1)^2 \\ = (2x+1)(2x+1) \end{array}$$

$$4x^2 + 2x + 2x + 1$$

$$4x^2 + 4x + 1$$

Q10

a) $4x^2 + 18x$

HCF = $2x$

$= 2x(2x+9)$

b) $4xy + 8x^2 - 8x^3y$

HCF = $4x$

$4x(y + 2x - 2x^2y)$

Q11 a) $\underline{x^2 + x + 4x + 4}$
 $x(x+1) + 4(x+1)$
 $= \underline{(x+1)(x+4)}$

b) $u(v-2) + 7(v-2)$
HCF = $(v-2)$
 $= \underline{(v-2)(u+7)}$

c) $x^2 + 7x + 10$

$x \begin{matrix} \times & 1 & 2 \\ & 10 & 5 \end{matrix}$

$= \underline{(x+2)(x+5)}$

d) $x^2 + 3x - 10$

$x \begin{matrix} \times & 2 & -2 \\ & -5 & 5 \end{matrix}$

$= \underline{(x-2)(x+5)}$

e) $x^2 - 9$
 $\sqrt{x^2} = x \quad \sqrt{9} = 3$
 $= \underline{(x+3)(x-3)}$

f) $1 - y^2$
 $= \underline{(1+y)(1-y)}$

Q12 a) $a^2 - 16 = 0$

$(a+4)(a-4) = 0$

$\therefore a+4=0 \text{ or } a-4=0$

$\underline{a=-4} \quad \underline{a=4}$

$\therefore a = 4 \text{ or } -4$

b) $x^2 + 2x + 1 = 0$

$\cancel{x} \begin{matrix} \times & 1 \\ & 1 \end{matrix}$

$(x+1)(x+1) = 0$

$\therefore x+1=0 \text{ or } x+1=0$
↖ same ↗

$\underline{x = -1}$

Q13

a) $\log_5 625 = \underline{4}$

b) $\log_{10} 2 + \log_{10} 5$
 $= \log_{10} (2 \times 5) = \log_{10} 10$
 $= \underline{1}$

Q14. a) $4x < 8$

$\underline{x < 2}$

b) $-2x \geq 16$ c) $x+5 < 21$

$x \overset{\text{sign flips}}{\leq} -8$

$\underline{x < 16}$

Q15. a) $\frac{x}{5} + \frac{2x}{5}$

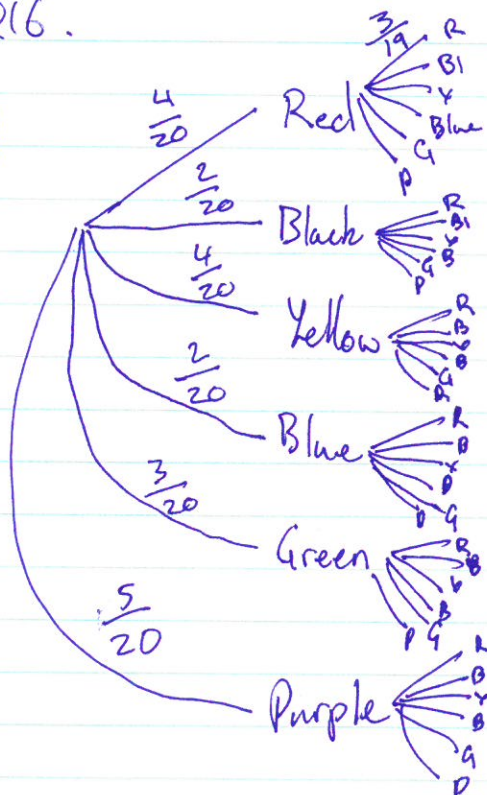
$= \frac{3x}{5}$

b) $\frac{7x}{4} - \frac{3x}{4}$

$= \frac{4x}{4} = \underline{x}$

Q16.

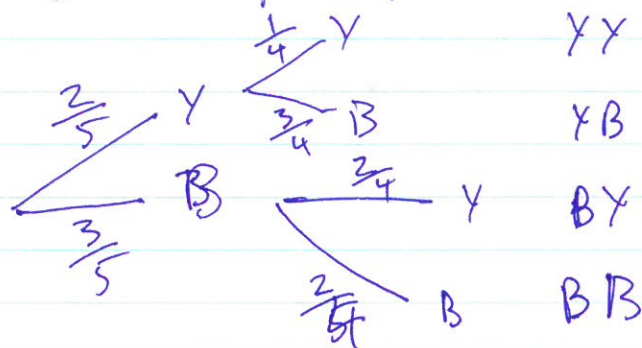
a)



b) $P(2 \text{ Red}) = \frac{4}{20} \times \frac{3}{19} = \frac{12}{380}$
 $= \frac{3}{95}$

Q17

a) Sample space



b) $P(2 \text{ yellow}) = \frac{2}{5} \times \frac{1}{4} = \frac{2}{20}$

c) $P(\text{yellow} + \text{black}) = \frac{2}{5} \times \frac{3}{4} + \frac{3}{5} \times \frac{2}{4}$
 $= \frac{6}{20} + \frac{6}{20} = \frac{12}{20}$

d) $P(\text{Yellow given first is black})$

$= \frac{2}{4} = \frac{1}{2}$

Q18



a) only dogs 30

b) cats 15

c) only cats 5